DERIVATIONS

Unit 3: Supersonic Flow

Prandtl–Meyer Expansion · Shock–Expansion Theory

Compiled from the course PDFs in this repository:
Unit 3_A to Unit 3_E

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Common Notation

Symbols used throughout. Each derivation also lists the meaning of every symbol under its formulas (“where, …”).

γ	=	ratio of specific heats, c_p/c_v (1.4 for air)
R	=	specific gas constant (287 J/kg·K for air)
c_p	=	specific heat at constant pressure (1005 J/kg·K for air)
$p,\ T,\ \rho$	=	static pressure, static temperature, density
$p_0,\ T_0,\ \rho_0$	=	stagnation (total) pressure, temperature, density
$p^*,\ T^*,\ \rho^*$	=	values at the sonic ($M = 1$) condition
$u,\ V$	=	flow velocity
a	=	speed of sound, $\sqrt{\gamma RT}$
M	=	Mach number, V/a
$A,\ A^*$	=	local area; sonic throat area
μ	=	Mach angle, $\sin^{-1}(1/M)$
$\theta,\ \beta$	=	flow deflection angle; oblique shock wave angle
$\nu(M)$	=	Prandtl–Meyer function
$\alpha,\ \varepsilon$	=	angle of attack; wedge half-angle of the diamond airfoil
$c,\ t,\ l$	=	chord, thickness, side length of an airfoil
$L,\ D$	=	lift and drag per unit span
$C_l,\ C_d$	=	lift and drag coefficients
q_∞	=	free-stream dynamic pressure, $\frac{\gamma}{2}pM^2$

PART A

Prandtl–Meyer Expansion Waves

Source: Unit 3_A.pdf, Unit 3_B.pdf

A1. Differential Relation Inside an Expansion Wave

Source: Unit 3_A.pdf

Setup: A single Mach wave is produced by an infinitesimal flow deflection $d\theta$. It makes Mach angle μ with the upstream flow. The velocity changes from V to $V + dV$ across it; the velocity component *along* the wave is unchanged.

Step 1: Equal tangential components (from similar triangles ADB and BEF, $BD = BE$)

$$V\cos\mu = (V + dV)\cos(\mu + d\theta)$$

where,

V = upstream velocity

$V + dV$ = velocity after the Mach wave

μ = Mach angle

$d\theta$ = infinitesimal flow deflection

Step 2: Rearrange and expand $\cos(\mu + d\theta)$

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos(\mu + d\theta)} = \frac{\cos\mu}{\cos\mu\cos d\theta - \sin\mu\sin d\theta}$$

Step 3: Small-angle assumption: $\sin d\theta \approx d\theta$, $\cos d\theta \approx 1$

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu - d\theta\sin\mu} = \frac{1}{1 - d\theta\tan\mu}$$

Step 4: Series expansion $\dfrac{1}{1-x} = 1 + x + x^2 + \cdots\ (x < 1)$, ignoring second-order and higher terms

$$1 + \frac{dV}{V} = 1 + d\theta\tan\mu \;\Rightarrow\; d\theta = \frac{dV/V}{\tan\mu}$$

where,

x = $d\theta\tan\mu$ (very small)

Step 5: Mach angle: $\sin\mu = 1/M$ (right triangle with hypotenuse M, opposite side 1, adjacent side $\sqrt{M^2-1}$)

$$\tan\mu = \frac{1}{\sqrt{M^2 - 1}}$$

where,

M = Mach number ahead of the wave

$$\boxed{d\theta = \sqrt{M^2 - 1}\,\frac{dV}{V}} \qquad \cdots (1)$$

where,

$d\theta$ = flow turning angle through the Mach wave

dV/V = fractional change of velocity

> **Note:** Equation (1) is the differential equation governing the flow inside an expansion fan. Since $ds = 0$ for each Mach wave, the expansion is isentropic: p_0 and T_0 stay constant.

A2. Prandtl–Meyer Function $\nu(M)$

Source: Unit 3_A.pdf

To find: the total turning angle θ when the Mach number rises from M_1 to M_2 in an expansion fan.

Step 1: Integrate eqn (1) from region 1 ($\theta = 0$, M_1) to region 2 (θ, M_2)

$$\int_0^{\theta} d\theta = \theta = \int_{M_1}^{M_2} \sqrt{M^2 - 1}\,\frac{dV}{V} \qquad \cdots (2)$$

where,

θ = total flow deflection angle

$M_1,\ M_2$ = Mach numbers upstream and downstream of the fan

Step 2: Express dV/V in terms of dM. From $V = Ma$ (take logs and differentiate)

$$\ln V = \ln M + \ln a \ \Rightarrow\ \frac{dV}{V} = \frac{dM}{M} + \frac{da}{a} \qquad \cdots (3)$$

where,

a = local speed of sound

Step 3: Isentropic relation for a

$$\left(\frac{a_0}{a}\right)^2 = \frac{T_0}{T} = 1 + \frac{\gamma - 1}{2}M^2 \ \Rightarrow\ a = a_0\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1/2}$$

where,

a_0 = stagnation speed of sound

$$\frac{da}{a} = -\frac{\gamma - 1}{2}\,M\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1} dM \qquad \cdots (4)$$

Step 4: Put (4) in (3)

$$\frac{dV}{V} = \frac{dM}{M}\left[1 - \frac{\frac{\gamma-1}{2}M^2}{1 + \frac{\gamma-1}{2}M^2}\right] = \frac{1}{1 + \frac{\gamma-1}{2}M^2}\,\frac{dM}{M} \qquad \cdots (5)$$

Step 5: Put (5) in (2) and integrate

$$\theta = \int_{M_1}^{M_2} \frac{\sqrt{M^2 - 1}}{1 + \frac{\gamma-1}{2}M^2}\,\frac{dM}{M} = \left[\sqrt{\tfrac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\tfrac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}\right]_{M_1}^{M_2}$$

$$\boxed{\theta = \nu(M_2) - \nu(M_1)}$$

where,

$\nu(M)$ = Prandtl–Meyer function

$$\boxed{\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\ \tan^{-1}\sqrt{\frac{\gamma-1}{\gamma+1}\,(M^2-1)}\ -\ \tan^{-1}\sqrt{M^2-1}}$$

where,

γ = ratio of specific heats

$\tan^{-1}$ = inverse tangent, result in radians/degrees

Step 6: Procedure to get the downstream state

1. For given M_1 read $\nu(M_1)$ from the Prandtl–Meyer table.
2. Compute $\nu(M_2) = \nu(M_1) + \theta$.
3. Read M_2 from the table for this $\nu(M_2)$.
4. Expansion is isentropic: $T_{02} = T_{01}$ and $p_{02} = p_{01}$.
5. From isentropic tables read $p_1/p_{01},\ T_1/T_{01}$ at M_1 and $p_2/p_{02},\ T_2/T_{02}$ at M_2.

$$\frac{p_2}{p_1} = \frac{p_2/p_{02}}{p_1/p_{01}}, \qquad \frac{T_2}{T_1} = \frac{T_2/T_{02}}{T_1/T_{01}}$$

where,

$p_0,\ T_0$ = stagnation pressure/temperature (same before and after the fan)

Step 7: Mach lines bounding the fan

$$\mu_1 = \sin^{-1}\frac{1}{M_1}\ \text{(forward Mach line)}, \qquad \mu_2 = \sin^{-1}\frac{1}{M_2}\ \text{(rearward Mach line, measured from the downstream flow)}$$

where,

$\mu_2 - \theta$ = angle of the rearward Mach line w.r.t. the upstream flow direction

A3. Maximum Turning Angle of a Prandtl–Meyer Expansion

Source: Unit 3_B.pdf

To find: the largest θ obtainable by expansion. From $\theta = \nu(M_2) - \nu(M_1)$, θ is largest when M_1 is smallest and M_2 is largest.

Step 1: Limits: $M_1 = 1$ (sonic start) and $M_2 = \infty$

$$\nu(M_1 = 1) = 0$$

where,

$\nu(1)$ = $\tan^{-1} 0 - \tan^{-1} 0 = 0$

$$\nu(M_2 \to \infty) = \sqrt{\tfrac{\gamma+1}{\gamma-1}} \cdot 90^\circ - 90^\circ$$

where,

$\tan^{-1} \infty$ = 90°

Step 2: Result

$$\boxed{\theta_{max} = \left(\sqrt{\tfrac{\gamma+1}{\gamma-1}} - 1\right) 90^\circ}$$

where,

θ_{max} = maximum possible turning angle

$$\gamma = 1.4: \quad \theta_{max} = (\sqrt{6} - 1)\, 90^\circ = 130.45^\circ$$

PART B

Shock–Expansion Theory: Supersonic Airfoils

Source: Unit 3_C.pdf, Unit 3_D.pdf, Unit 3_E.pdf

B1. Lift and Drag of a Flat Plate at Angle of Attack

Source: Unit 3_C.pdf

Setup: a thin flat plate of chord c at small angle of attack α in supersonic flow $M_1 > 1$ (p_1). The upper surface sees an expansion fan (region 2: $M_2 > M_1$, $p_2 < p_1$). The lower surface sees an oblique shock (region 3: $M_3 < M_1$, $p_3 > p_1$). Forces are per unit span.

Step 1: Resultant pressure force (normal to the plate)

$$R = p_3 c - p_2 c = (p_3 - p_2)\, c$$

where,

R = resultant force normal to plate

p_2 = pressure on upper surface

p_3 = pressure on lower surface

c = plate chord

Step 2: Resolve perpendicular and parallel to the free stream

$$\boxed{L = R\cos\alpha = (p_3 - p_2)\, c\cos\alpha}$$

where,

L = lift (perpendicular to free stream)

α = angle of attack

$$\boxed{D = R\sin\alpha = (p_3 - p_2)\, c\sin\alpha}$$

where,

D = drag (parallel to free stream)

B2. Dynamic Pressure in Terms of Mach Number

Source: Unit 3_D.pdf

Step 1: Start from the definition and multiply and divide by γp_∞

$$q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 = \frac{1}{2}\,\frac{\gamma p_\infty}{\gamma p_\infty}\,\rho_\infty V_\infty^2 = \frac{1}{2}\,\gamma p_\infty\,\frac{\rho_\infty}{\gamma p_\infty} V_\infty^2$$

where,

q_∞ = free-stream dynamic pressure

$\rho_\infty,\ p_\infty,\ V_\infty$ = free-stream density, pressure, velocity

Step 2: Use $a_\infty = \sqrt{\gamma p_\infty/\rho_\infty}$

$$q_\infty = \frac{1}{2}\gamma p_\infty \frac{V_\infty^2}{a_\infty^2} \;\Rightarrow\; \boxed{q_\infty = \frac{\gamma}{2}\, p_\infty M_\infty^2}$$

where,

a_∞ = free-stream speed of sound

M_∞ = free-stream Mach number

B3. Lift and Drag Coefficients of a Flat Plate

Source: Unit 3_D.pdf

Step 1: Definitions, with $q_1 = \frac{\gamma}{2} p_1 M_1^2$ and reference area $= c$ (per unit span)

$$C_l = \frac{L}{q_1 c}, \qquad C_d = \frac{D}{q_1 c}$$

where,

$C_l,\ C_d$ = lift and drag coefficients

q_1 = dynamic pressure of the free stream

Step 2: Substitute L, D from B1

$$C_l = \frac{(p_3 - p_2)\, c \cos\alpha}{\frac{\gamma}{2} p_1 M_1^2\, c}$$

$$\boxed{C_l = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\cos\alpha}$$

where,

p_2/p_1 = from expansion fan (1→2), via $\nu(M)$ tables

p_3/p_1 = from oblique shock (1→3), via θ–β–M chart and normal shock tables

$$\boxed{C_d = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\sin\alpha}$$

Step 3: Pressure ratios

$$\frac{p_2}{p_1} = \frac{p_2}{p_{02}} \cdot \frac{p_{02}}{p_{01}} \cdot \frac{p_{01}}{p_1} = \frac{p_2/p_{02}}{p_1/p_{01}} \quad (p_{01} = p_{02}, \text{isentropic expansion})$$

$\frac{p_3}{p_1}$ from oblique shock: $M_{n1} = M_1 \sin\beta$

where,

β = oblique shock wave angle

M_{n1} = Mach number component normal to the shock

B4. Diamond Airfoil at Zero Angle of Attack ($\alpha = 0$)

Source: Unit 3_C.pdf

Setup: diamond airfoil with chord *c*, thickness t, side length l and wedge half-angle ε. At $\alpha = 0$ the flow is symmetric. Front faces: oblique shock, pressure $p_2 > p_1$. Rear faces: expansion fan, pressure $p_3 < p_2$. (By symmetry $p_4 = p_2$, $p_5 = p_3$.) Geometry: $l \sin\varepsilon = t/2$, $l \cos\varepsilon = c/2$.

Step 1: Force on each face = pressure × side length ($p\, l$), normal to the face. Sum the components

$$L = -p_2 l \cos\varepsilon + p_2 l \cos\varepsilon - p_3 l \cos\varepsilon + p_3 l \cos\varepsilon \;\Rightarrow\; \boxed{L = 0}$$

where,

L = lift per unit span (perpendicular to free stream)

$p_2 l$ = force on a front face (upper or lower)

$p_3 l$ = force on a rear face (upper or lower)

ε = wedge half-angle

$$D = p_2 l \sin\varepsilon - p_3 l \sin\varepsilon + p_2 l \sin\varepsilon - p_3 l \sin\varepsilon = 2(p_2 - p_3)\, l \sin\varepsilon$$

where,

D = wave drag per unit span (parallel to free stream)

Step 2: Use $l \sin\varepsilon = t/2$

$$\boxed{D = (p_2 - p_3)\, t}$$

where,

t = maximum thickness of the airfoil

> **Note:** In inviscid *incompressible* flow the drag is zero (d'Alembert's paradox). In inviscid supersonic flow the drag is finite: **wave drag**. It arises purely from shock waves (entropy rise and loss of total pressure).

B5. Diamond Airfoil at Angle of Attack $\alpha > \varepsilon$: Lift and Drag Coefficients

Source: Unit 3_E.pdf

Wave pattern (all four faces see a different pressure):

Face	Wave	Turning angle θ
1 → 2 (upper front)	expansion fan	$\theta_2 = \alpha - \varepsilon$
2 → 3 (upper rear)	expansion fan	$\theta_3 = 2\varepsilon$
1 → 4 (lower front)	oblique shock	$\theta_4 = \alpha + \varepsilon$
4 → 5 (lower rear)	expansion fan	$\theta_5 = 2\varepsilon$

Step 1: Normal force N (perpendicular to the chord line); $l\cos\varepsilon = c/2$

$$N = -p_2 l\cos\varepsilon - p_3 l\cos\varepsilon + p_4 l\cos\varepsilon + p_5 l\cos\varepsilon$$

where,

N = force component normal to chord

p_2, p_3 = pressures on upper front/rear faces

p_4, p_5 = pressures on lower front/rear faces

$$\boxed{N = (p_4 + p_5 - p_2 - p_3)\,\frac{c}{2}}$$

where,

c = chord length

Step 2: Axial force A (along the chord line); $l\sin\varepsilon = t/2$

$$A = p_2 l\sin\varepsilon + p_4 l\sin\varepsilon - p_3 l\sin\varepsilon - p_5 l\sin\varepsilon$$

where,

A = force component along chord

$$\boxed{A = (p_2 + p_4 - p_3 - p_5)\,\frac{t}{2}}$$

where,

t = maximum thickness

Step 3: Rotate to the wind axes (free-stream direction)

$$\boxed{L = N\cos\alpha - A\sin\alpha}, \qquad \boxed{D = N\sin\alpha + A\cos\alpha}$$

where,

L, D = lift and drag per unit span

α = angle of attack

Step 4: Coefficients, with $q_1 = \frac{\gamma}{2}p_1 M_1^2$ and reference area c

$$C_l = \frac{L}{q_1 c} = \frac{2}{\gamma M_1^2 c}\left[\left(\tfrac{p_4}{p_1} + \tfrac{p_5}{p_1} - \tfrac{p_2}{p_1} - \tfrac{p_3}{p_1}\right)\frac{c}{2}\cos\alpha - \left(\tfrac{p_2}{p_1} + \tfrac{p_4}{p_1} - \tfrac{p_3}{p_1} - \tfrac{p_5}{p_1}\right)\frac{t}{2}\sin\alpha\right]$$

$$\boxed{C_l = \frac{1}{\gamma M_1^2}\left[\left(\tfrac{p_4}{p_1} + \tfrac{p_5}{p_1} - \tfrac{p_2}{p_1} - \tfrac{p_3}{p_1}\right)\cos\alpha - \left(\tfrac{p_2}{p_1} + \tfrac{p_4}{p_1} - \tfrac{p_3}{p_1} - \tfrac{p_5}{p_1}\right)\frac{t}{c}\sin\alpha\right]}$$

where,

t/c = thickness ratio, $= \tan\varepsilon$

p_i/p_1 = pressure ratio of face i to free stream

$$\boxed{C_d = \frac{1}{\gamma M_1^2}\left[\left(\tfrac{p_4}{p_1} + \tfrac{p_5}{p_1} - \tfrac{p_2}{p_1} - \tfrac{p_3}{p_1}\right)\sin\alpha + \left(\tfrac{p_2}{p_1} + \tfrac{p_4}{p_1} - \tfrac{p_3}{p_1} - \tfrac{p_5}{p_1}\right)\frac{t}{c}\cos\alpha\right]}$$

Step 5: Each pressure ratio is found with the shock/expansion tools

$$\frac{p_2}{p_1} = \frac{p_2/p_{02}}{p_1/p_{01}}, \quad \frac{p_3}{p_1} = \frac{p_3/p_{03}}{p_1/p_{01}}\ (p_{01} = p_{02} = p_{03}), \quad \frac{p_5}{p_1} = \frac{p_5}{p_{05}}\cdot\frac{p_{04}}{p_{01}}\cdot\frac{p_{01}}{p_1}\ (p_{04} = p_{05})$$

where,

p_{0i} = stagnation pressure in region i

p_{04}/p_{01} = stagnation-pressure ratio across the oblique shock (from normal shock tables with M_{n1})

$$\nu(M_2) = \nu(M_1) + \theta_2, \quad \nu(M_3) = \nu(M_2) + \theta_3, \quad M_4 = \frac{M_{n4}}{\sin(\beta_4 - \theta_4)}, \quad \nu(M_5) = \nu(M_4) + \theta_5$$

where,

β_4 = shock angle for the lower front face

M_{n4} = normal Mach number downstream of the shock